

# Beautiful Autumn Live Wallpape

com.amaxlwps.beautifulautumn

|            |                 |
|------------|-----------------|
| Developer  | Amax LWPS       |
| Genre      | PERSONALIZATION |
| Installs   | 100,000+        |
| Audit Date | 2026-04-26      |

Overall Risk Score

0.47

MEDIUM

2

Consistent

3

Mismatches

5

Over-disclosure

2

Undeclared

Research prototype · Ongoing research | Policy: <https://amaxsoftware.com/privacy-policy/>

## Executive Summary

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**Beautiful Autumn Live Wallpaper** declares 4 data type(s) collected and 4 shared in its Play Data Safety label, across 12 data type(s) analyzed. Our heterogeneous GNN audit model identifies **2 potential undeclared collection(s)** — data types implied by embedded SDK signals but absent from both label and policy — and **3 policy-vs-label mismatch(es)**. Overall risk score: **0.47 (MEDIUM)**.

### Top discrepancies by risk:

- **installed\_apps**: Undeclared Collection (confidence 0.97)
- **crash\_logs**: Undeclared Collection (confidence 0.96)
- **app\_interactions**: Policy Label Mismatch (confidence 1.00)

### Class definitions:

**CONSISTENT**: Label, policy, and SDK signals agree.

**POLICY\_LABEL\_MISMATCH**: Label discloses data not mentioned in policy.

**OVER\_DISCLOSURE**: Policy mentions data not declared in label.

**UNDECLARED\_COLLECTION**: SDK signals collection undisclosed in label and policy.

## Discrepancy Table

Sorted by risk class (undeclared collection first). Y/N: label=data-safety label, policy=policy text, sdk=SDK signal.

| Data Type            | Class                 | Conf. | Evidence                            |
|----------------------|-----------------------|-------|-------------------------------------|
| installed_apps       | Undeclared Collection | 0.97  | label:N policy:N sdk:Y (1, 2, 4 +1) |
| crash_logs           | Undeclared Collection | 0.96  | label:N policy:N sdk:Y (1, 2, 4 +1) |
| app_interactions     | Policy Label Mismatch | 1.00  | label:Y policy:N sdk:N (1, 2, 4 +1) |
| diagnostics          | Policy Label Mismatch | 0.99  | label:Y policy:N sdk:Y (1, 2, 4 +1) |
| approximate_location | Policy Label Mismatch | 0.98  | label:Y policy:N sdk:Y (1, 2, 4 +1) |
| date_of_birth        | Over Disclosure       | 0.97  | label:N policy:Y sdk:N (1, 2, 4 +1) |
| files_and_docs       | Over Disclosure       | 0.96  | label:N policy:Y sdk:N (1, 2, 4 +1) |
| email_address        | Over Disclosure       | 0.95  | label:N policy:Y sdk:N (1, 2, 4 +1) |
| phone_number         | Over Disclosure       | 0.95  | label:N policy:Y sdk:N (1, 2, 4 +1) |
| name                 | Over Disclosure       | 0.92  | label:N policy:Y sdk:N (1, 2, 4 +1) |
| device_id            | Consistent            | 1.00  | label:Y policy:Y sdk:Y (1, 2, 4 +1) |
| advertising_id       | Consistent            | 0.99  | label:N policy:Y sdk:Y (1, 2, 4 +1) |

## Evidence Trails

Detailed evidence for UNDECLARED\_COLLECTION or confidence > 0.85 non-CONSISTENT findings.

### Discrepancy: installed\_apps — Undeclared Collection (confidence 0.97)

Label declares collection: **No**

Label declares sharing: **No**

Policy text mentions: **No**

SDK collection signals: **Yes** — 1, 2, 4, 35

Recommended action: Add 'installed\_apps' to the Play Data Safety label under 'Data Collected' and update the privacy policy to disclose this collection.

### Discrepancy: crash\_logs — Undeclared Collection (confidence 0.96)

Label declares collection: **No**

Label declares sharing: **No**

Policy text mentions: **No**

SDK collection signals: **Yes** — 1, 2, 4, 35

Recommended action: Add 'crash\_logs' to the Play Data Safety label under 'Data Collected' and update the privacy policy to disclose this collection.

### Discrepancy: app\_interactions — Policy Label Mismatch (confidence 1.00)

Label declares collection: **Yes**

Label declares sharing: **Yes**

Policy text mentions: **No**

SDK collection signals: **No** — 1, 2, 4, 35

Recommended action: Update the privacy policy to disclose 'app\_interactions' collection, aligning it with the Play Data Safety label.

### Discrepancy: diagnostics — Policy Label Mismatch (confidence 0.99)

Label declares collection: **Yes**

Label declares sharing: **Yes**

Policy text mentions: **No**

SDK collection signals: **Yes** — 1, 2, 4, 35

Recommended action: Update the privacy policy to disclose 'diagnostics' collection, aligning it with the Play Data Safety label.

#### Discrepancy: `approximate_location` — **Policy Label Mismatch** (confidence 0.98)

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Label declares collection: **Yes**

Label declares sharing: **Yes**

Policy text mentions: **No**

SDK collection signals: **Yes** — 1, 2, 4, 35

Recommended action: Update the privacy policy to disclose '`approximate_location`' collection, aligning it with the Play Data Safety label.

#### Discrepancy: `date_of_birth` — **Over Disclosure** (confidence 0.97)

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Label declares collection: **No**

Label declares sharing: **No**

Policy text mentions: **Yes**

SDK collection signals: **No** — 1, 2, 4, 35

Recommended action: Review whether '`date_of_birth`' is genuinely collected. If yes, add to label; if no, remove from policy.

#### Discrepancy: `files_and_docs` — **Over Disclosure** (confidence 0.96)

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Label declares collection: **No**

Label declares sharing: **No**

Policy text mentions: **Yes**

SDK collection signals: **No** — 1, 2, 4, 35

Recommended action: Review whether '`files_and_docs`' is genuinely collected. If yes, add to label; if no, remove from policy.

#### Discrepancy: `email_address` — **Over Disclosure** (confidence 0.95)

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Label declares collection: **No**

Label declares sharing: **No**

Policy text mentions: **Yes**

SDK collection signals: **No** — 1, 2, 4, 35

Recommended action: Review whether '`email_address`' is genuinely collected. If yes, add to label; if no, remove from policy.

#### Discrepancy: `phone_number` — **Over Disclosure** (confidence 0.95)

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Label declares collection: **No**

Label declares sharing: **No**

Policy text mentions: **Yes**

SDK collection signals: **No** — 1, 2, 4, 35

Recommended action: Review whether '`phone_number`' is genuinely collected. If yes, add to label; if no, remove from policy.

### Discrepancy: name — **Over Disclosure** (confidence 0.92)

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Label declares collection: **No**

Label declares sharing: **No**

Policy text mentions: **Yes**

SDK collection signals: **No** — 1, 2, 4, 35

Recommended action: Review whether 'name' is genuinely collected. If yes, add to label; if no, remove from policy.

## Methodology

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PolicyGraphAudit-RT represents each Android app as a tri-partite heterogeneous graph connecting three layers: **Privacy Policy** text (OPP-115 segment classification), **Play Data Safety Labels** (declared data types and purposes), and **Runtime Evidence** (inferred SDK trackers via Exodus Privacy). A 2-layer HeteroConv R-GCN encodes all three layers jointly; an MLP classifier head predicts discrepancy classes for each (App, DataType) pair.

The model was trained under an edge-masking protocol (30% of label-determining edges removed) to prevent circularity. It achieves **macro F1 = 0.956** on 39 held-out apps (UNDECLARED\_COLLECTION F1 = 0.974), trained on 252 apps and 3,202 labeled pairs. Model card: [https://github.com/PolicyGraphAudit-RT/reports/blob/main/reports/m5\\_model\\_card.md](https://github.com/PolicyGraphAudit-RT/reports/blob/main/reports/m5_model_card.md).

Known limitations: (1) Privacy labels are from a 2022 Play Store snapshot. (2) SDK presence is inferred via category-level priors, not measured from APK bytecode. (3) Policy classifier uses MiniLM + logistic regression (OPP-115 macro F1 = 0.70). (4) Labels are rule-derived weak supervision, not human-annotated. (5) English-language Android apps only.

## Disclaimer

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This report is produced by a research prototype and is provided for informational and research purposes only. It does not constitute legal advice or a regulatory determination. Findings are based on automated weak-supervision labels from publicly available data and have not been reviewed by a privacy legal professional. Research prototype · Ongoing research.